

Waste Hotspot Detection, Priority Ranking and What-If Simulation in a Software-Based Community Digital Twin

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ABSTRACT

Communities usually receive information about waste, drainage, flooding and water shortages through separate channels, so administrators cannot easily see where problems cluster or which one to handle first. This study presents three decision-support components of CommunityTwin, an AI-powered digital twin of a rural or urban community: waste hotspot detection from geotagged citizen reports, a weighted priority engine that ranks locations across water, flood and waste risks, and a what-if simulation module for planning scenarios. On simulated data, density-based clustering (DBSCAN) recovered all four true hotspots with a recall of 0.994 and precision of 0.940, and a sensitivity analysis showed that results depend strongly on the neighbourhood radius. The priority score shared only three of five locations with a ranking by the single largest risk, showing that combining hazards and exposure changes escalation order. A drainage-improvement and waste-clearing scenario reduced mean predicted flood probability by 40.3%, and a 50% rainfall increase raised it by about 41%. The evaluation uses synthetic data, so the results demonstrate feasibility rather than field accuracy.

1. Introduction

Municipal and village administrators routinely receive information about waste accumulation, blocked drains, flooding and water shortages from different channels: complaint registers, phone calls, field visits and, increasingly, mobile applications. Because these channels are not integrated, administrators find it difficult to see where problems cluster, which problem is most urgent, and what an intervention would achieve. A pile of uncollected waste, a blocked drain and a flooded street may be recorded by three different offices in three different formats, even though the first may be the cause of the second and the second of the third. Digital twin (DT) technology, defined as a virtual replica that is continuously synchronised with its physical counterpart and used for analysis and simulation (Grieves and Vickers 2017; Fuller et al. 2020), offers a natural framework for such integration.

Smart-city DTs have mostly been built from detailed BIM-GIS models, which are expensive to build and heavy to serve on the web (Jin et al. 2024). AI-enabled water management frameworks have shown the value of ensemble learning on sensor data (Rokaya Mahmoud et al. 2025), but they do not address waste, do not integrate citizen reports and do not provide decision-level prioritisation. For a rural panchayat or an urban ward office with a small technical staff, what is needed is not a high-fidelity model of every building but a lightweight, transparent system that answers three practical questions: *where* are problems concentrated, *which* location should be handled first, and *what* would a given intervention achieve?

This paper focuses on three decision-support components of the CommunityTwin platform that complement the prediction modules for water shortage and flood risk described elsewhere: (1) waste hotspot detection from geotagged citizen reports and images, (2) a priority engine that ranks locations across hazards, and (3) a what-if simulation facility. Each component corresponds to one of the three questions above. Together they turn the outputs of the prediction modules and the stream of citizen reports into a ranked, explainable list of actions.

The study is organised around three research questions:

- **RQ1.** Can density-based clustering separate genuine waste hotspots from scattered, one-off citizen complaints, and how sensitive is it to its two parameters?
- **RQ2.** Does a transparent weighted score that combines water, flood and waste risk with exposure change the order in which locations are escalated compared with ranking by the largest single risk, and how stable is the ranking with respect to the weights?
- **RQ3.** Can a what-if module built on the trained prediction models give directionally sensible estimates of the effect of planned interventions?

The contributions are: a spatial pipeline that converts raw citizen reports into verified hotspots; a transparent weighted priority score with a sensitivity analysis of its weights; a what-if module for planning scenarios; and an empirical evaluation on simulated data with clearly stated limitations. All data are synthetic, so the results demonstrate feasibility and expose sensitivities rather than establishing field accuracy.

The rest of the paper is structured as follows. Section 2 reviews background and related work. Section 3 describes the proposed methods. Section 4 reports the experiments. Section 5 discusses the findings and their practical implications, Section 6 sets out limitations and threats to validity, and Section 7 concludes.

2. Background and Related Work

2.1 Digital twins for communities

Tao et al. (2019) describe DTs as a fusion of physical and virtual data, and Fuller et al. (2020) point out open issues in data quality, interoperability and security. Jin et al. (2024) propose a lightweighting process for smart-city DTs so that numerous real-time services can run on the web. CommunityTwin follows this direction by representing each ward or village as a compact state vector rather than a detailed geometry. A compact state has two advantages for the components studied here. It makes the data requirements modest, so that a community can start with whatever it has, and it makes every number on the dashboard traceable to a small set of named inputs, which is important when recommendations are to be explained to officials and residents.

2.2 Spatial clustering of citizen reports

Density-based clustering identifies arbitrarily shaped dense regions and labels isolated points as noise (Ester et al. 1996), which makes it suitable for separating genuine waste hotspots from scattered one-off complaints. DBSCAN requires two parameters: a neighbourhood radius *eps* and a minimum number of points *minPts*. A point is a *core point* if at least *minPts* points (including itself) lie within distance *eps*; clusters are formed by linking core points whose neighbourhoods overlap, together with the *border points* within reach of a core point; all remaining points are *noise*. Unlike centroid-based methods such as *k*-means, DBSCAN does not require the number of clusters to be known in advance, does not force every point into a cluster, and can follow irregular shapes such as a hotspot stretched along a road. Its main weakness is that a single global density threshold is difficult to choose when hotspots differ in density, as they do when a dense urban market and a sparse village outskirts are in the same map. Hierarchical variants (Campello et al. 2013) remove the need for a single density threshold and are a candidate for future work.

2.3 Multi-criteria prioritisation

Weighted multi-criteria methods such as the analytic hierarchy process (Saaty 1980) are widely used to rank alternatives when criteria differ in importance. They share a structure: criteria are scored on a common scale, weights express their relative importance, and a weighted aggregate yields a ranking. The strengths of this approach are transparency and the possibility of negotiation, since stakeholders can discuss weights in natural language. Its weaknesses are that weights are subjective, that a weighted sum allows a high score on one criterion to compensate for a low score on another, and that rankings can change when weights change. We use a simple weighted sum that administrators can inspect and modify, and we quantify the last weakness directly through a sensitivity analysis.

2.4 Scenario analysis in decision support

Scenario or what-if analysis asks how an output would change if inputs were different. In engineering practice such questions are answered with process-based simulators calibrated to a system. In data-driven decision support they are often answered by re-evaluating a statistical model on altered inputs. The latter is fast and requires no physical model, but it inherits the limitation that a statistical model encodes associations observed in historical data, not causal mechanisms. The estimate for an intervention that has no precedent in the data is therefore an extrapolation. We adopt the data-driven approach for its simplicity and are explicit about its status as an indicator of direction and approximate magnitude.

2.5 Citizen reports and image screening

Geotagged citizen reports turn residents into distributed sensors, which is valuable where physical sensing is sparse, but they are uneven in coverage, prone to duplicates and sometimes unreliable. Attaching a photograph allows a lightweight image classifier to screen out reports that do not show waste. The pipeline in this paper includes such a screening stage in its design but, as stated below, does not evaluate it.

2.6 Positioning

Table 1 summarises how CommunityTwin differs from the studies discussed above. The surveyed works either review the DT concept, address a single service or asset, or rely on detailed geometric or dense sensor data. CommunityTwin addresses several hazards jointly, works from sparse and simulated data, and treats prioritisation and what-if analysis as first-class functions.

Table 1. Comparison of CommunityTwin with related studies.

Study	Main focus	Data basis	Multi-hazard	Priority / what-if
Fuller et al. (2020)	DT survey	Literature	N/A	N/A
Tao et al. (2019)	DT in industry	Literature	N/A	N/A
Jin et al. (2024)	Lightweight smart-city DT	BIM-GIS	No	Not a focus
Rokaya Mahmoud et al. (2025)	Irrigation, leak detection	IoT sensors	No	Not a focus
CommunityTwin	Community DT	Sim. sensors, reports, images, weather, GIS	Yes	Yes

3. Proposed Methods

3.1 System overview

Fig. 1 shows the overall pipeline. Data from six sources (community data, simulated sensors, citizen reports, images, weather and GIS) enter a collection layer and are stored in a database. The community digital twin holds the state of each ward or village. AI/ML analysis produces three streams of risk information (water risk, flood risk and waste detection), which are combined by the priority engine into recommended actions and presented on a dashboard that also hosts the what-if simulation. This paper concentrates on the lower half of the pipeline: waste analysis, the priority engine, recommended actions and the what-if simulation module.

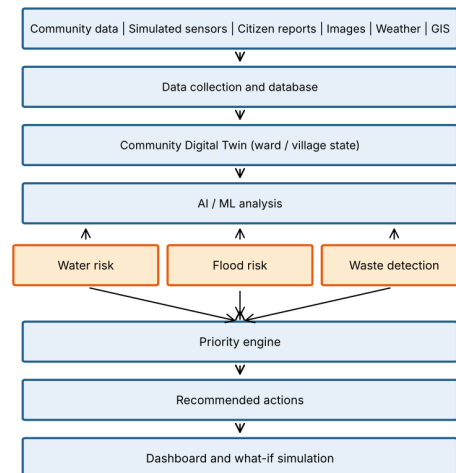


Fig. 1. Overall CommunityTwin pipeline from community data to dashboard and what-if simulation.

3.2 Methodology workflow

Each component follows the workflow in Fig. 2: data acquisition and validation (duplicate removal, coordinate checks, range checks), feature engineering, training and cross-validation, evaluation, and deployment in the twin, where new reports and readings trigger re-computation. Thresholds and weights are stored as configuration rather than code so that administrators can adjust them. For the components studied here, ‘training’ applies to the flood model that supports the what-if module; the clustering step has no training phase, and the priority engine is a fixed scoring rule with configurable weights.

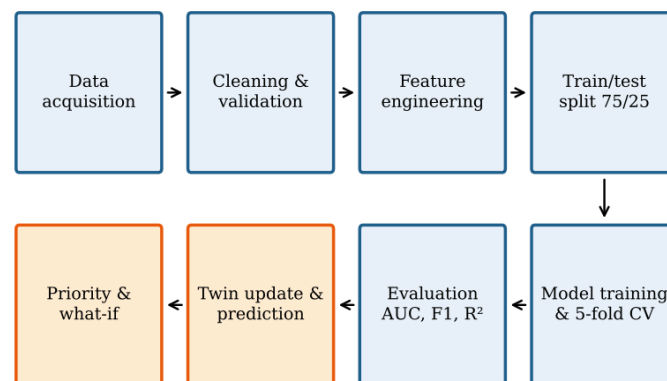


Fig. 2. Methodology workflow for training, evaluation and deployment.

The citizen report record used by the waste module is described in Table 2. Each record carries a location, a time stamp, a category and, optionally, an image. The *waste_flag* field stores the output of the image classifier. The status field allows the dashboard to distinguish open reports from those already assigned or resolved, so that resolved reports do not keep a hotspot alive indefinitely.

Table 2. Citizen report record used by the waste module.

Field	Type	Description
report_id	String	Unique identifier
lat, lon	Float	Geotag (normalised in experiments)
timestamp	Datetime	Time of submission
category	Enum	Waste / drainage / water supply / other
image	File (optional)	Photo submitted by the citizen
waste_flag	Boolean	Output of the image classifier (not evaluated here)
status	Enum	Open / assigned / resolved

3.3 Waste hotspot detection

Each citizen report contains a location, a timestamp and optionally an image. Images are classified by a lightweight image classifier into waste / no waste, and reports classified as no waste are removed. Valid report locations are then clustered by DBSCAN (Ester et al. 1996). A cluster with at least *minPts* reports within radius *eps* is declared a hotspot; reports outside any cluster are treated as noise. Each hotspot receives a severity score from its report count and age. In this paper the image-classification stage is described but not evaluated; the experiments evaluate the spatial stage only.

Distance and units. In the experiments, coordinates are normalised to the unit square and distances are Euclidean in these units, so *eps* = 0.03 corresponds to three per cent of the side of the map. In a deployment, latitude and longitude would be projected to a planar coordinate system (for example UTM) so that *eps* can be expressed in metres, which is more meaningful to administrators; a radius of a few tens of metres is a natural scale for a roadside waste heap, whereas a few hundred metres would merge neighbouring streets.

Severity. Severity is computed from the number of reports in the cluster and an age weight. The intention is that a cluster with many reports and a long-standing problem ranks above a recent cluster with few reports. The exact age weighting is a configuration choice; the experiments evaluate cluster detection rather than the severity formula, and the choice of weighting should be agreed with local staff. Algorithm 1 summarises the procedure.

Table 3. Algorithm 1: Waste hotspot detection and severity scoring.

Step	Operation
1	Input: reports $R = \{(lat, lon, time, image)\}$; parameters <i>eps</i> , <i>minPts</i>
2	Remove duplicates and reports with invalid coordinates
3	If an image is present, keep the report only if the classifier outputs waste
4	Run DBSCAN(<i>eps</i> , <i>minPts</i>) on the coordinates of the kept reports
5	For each cluster: centroid = mean location; severity = report count x age weight
6	Label points with cluster = -1 as noise and archive them
7	Output: hotspot list with centroid, radius, count and severity

Noise points are archived rather than discarded. A report that is isolated today may become part of a hotspot tomorrow as further reports arrive, and a re-run of DBSCAN on the archive plus new reports will then include it. Archiving also preserves an audit trail for reports that administrators wish to inspect individually.

3.4 Priority engine

For ward *i* the priority score is, with the weights and exposure terms defined below,

$$P_i = (w_w s_{w,i} + w_f s_{f,i} + w_g s_{g,i}) \times (0.6 + 0.25 pop_i + 0.15 vul_i) \quad (1)$$

with default weights $w_w = 0.35$, $w_f = 0.40$ and $w_g = 0.25$, reflecting the greater urgency of flood events. Here $s_{w,i}$, $s_{f,i}$ and $s_{g,i}$ are the normalised water, flood and waste risk scores of ward *i*, pop_i is its normalised population, and vul_i a vulnerability index. The score maps to four levels (low, medium, high, critical) that

trigger recommended actions such as tanker deployment, drain cleaning or additional waste collection (Table 4).

Properties of the score. Because the three hazard weights sum to one and the risk scores lie in $[0, 1]$, the hazard term lies in $[0, 1]$. The exposure term ranges from 0.6 for the smallest and least vulnerable ward to 1.0 for the largest and most vulnerable. Exposure can therefore raise or lower a ward's score by at most a factor of $1/0.6$, about 1.67, which means that it adjusts the ranking without overriding a large difference in hazard. The sum form is compensatory: a very high flood score can offset low water and waste scores, which is intended, since a severe hazard in one domain should still trigger action.

Worked example. The following example is hypothetical and is given only to illustrate the arithmetic. Ward A has $(s_w, s_f, s_g) = (0.5, 0.6, 0.3)$, $pop = 0.8$ and $vul = 0.5$. Its hazard term is $0.35 \times 0.5 + 0.40 \times 0.6 + 0.25 \times 0.3 = 0.490$ and its exposure multiplier is $0.6 + 0.25 \times 0.8 + 0.15 \times 0.5 = 0.875$, so $P_A = 0.429$. Ward B has $(0.2, 0.9, 0.1)$, $pop = 0.2$ and $vul = 0.1$, giving a hazard term of 0.455, a multiplier of 0.665 and $P_B = 0.303$. Ranking by the largest single risk would place B first (0.9 against 0.6), but the priority score places A first because A has risks in all three domains and affects more vulnerable people.

Table 4. Priority level and recommended action rules (levels by score percentile across wards).

Level	Score percentile	Recommended action
Critical	Top 10%	Immediate field team; notify ward officer
High	75th-90th	Schedule within 48 h (drain cleaning, tanker, extra pick-up)
Medium	40th-75th	Include in weekly work plan
Low	Below 40th	Monitor; no action

Levels are defined by percentile rather than by absolute score. This guarantees that a fixed share of wards is always escalated, which suits communities whose field capacity is limited, but it also means that the level is relative: in a community where all wards are at low risk, the top ward is still labelled 'critical'. The dashboard therefore shows the absolute score next to the level. With the 25 wards used in the experiments, the top 10% corresponds to two or three wards.

3.5 What-if simulation

A scenario is a set of changes to twin state variables (for example, drainage capacity +25%, waste blockage removed, or rainfall +50%). The trained prediction models are re-evaluated on the modified state and the change in predicted risk is reported. Because the models are statistical, the output indicates the direction and approximate magnitude of an effect rather than a physical simulation. The procedure has four steps: copy the current or representative state; apply the scenario's changes to the copy; evaluate the model on the modified copy; and report the change relative to the unmodified state. The live twin is never altered by a scenario. Table 5 lists the scenarios implemented in the prototype.

Table 5. What-if scenarios implemented in the prototype.

ID	Change to twin state	Module affected
S1	Drainage capacity +25%; waste-blockage index set to 0	Flood
S2	Rainfall (24 h and 72 h) +50%	Flood
S3 (planned)	Tank storage +20%; leakage -50%	Water

Scenario S1 couples the waste and flood domains: the waste-blockage index is derived from citizen reports and enters the flood model as an input, so clearing blockages can be represented as setting that index to zero. Scenario S2 represents a more severe rainfall regime and serves as a stress test. Scenario S3 is planned for the water module and is not evaluated here.

4. Experiments

4.1 Data and settings

All data are synthetic and generated with a fixed random seed. The waste data set consists of 1,500 geotagged reports: four true hotspots of 220 reports each (Gaussian spread, standard deviation 0.025 in normalised units) plus 620 uniformly distributed background reports. DBSCAN used $eps = 0.03$ and $minPts = 15$ (scikit-learn; Pedregosa et al. 2011). The priority engine was tested on 25 simulated wards. The what-if module used a random forest flood model (Breiman 2001) trained on 6,000 ward-day records.

Evaluation measures. Hotspot detection is evaluated at the level of reports. A report generated by one of the four Gaussian hotspots is a *true hotspot report*; a report from the uniform background is a *background report*. *Recall of hotspot reports* is the fraction of true hotspot reports assigned to a cluster. *Precision of hotspot reports* is the fraction of reports assigned to clusters that are true hotspot reports. *Background reports rejected as noise* is the fraction of background reports labelled as noise. The number of hotspots found is the number of clusters returned. There are $4 \times 220 = 880$ true hotspot reports and 620 background reports in the data set.

4.2 Waste hotspot detection results

Table 6. DBSCAN hotspot detection on simulated reports.

Metric	Value
Hotspots found (true = 4)	4
Recall of hotspot reports	0.994
Precision of hotspot reports	0.940
Background reports rejected as noise	0.910

DBSCAN recovered all four hotspots, with recall of 0.994 and precision of 0.940 (Table 6, Fig. 3). About 91.0% of background reports were rejected as noise; the remainder were absorbed because uniformly scattered points occasionally fall near a hotspot. Results depend strongly on the choice of eps and $minPts$, which were not tuned here and would need calibration on real data.

The counts implied by these rates help in interpreting them. With 880 true hotspot reports, a recall of 0.994 corresponds to roughly 875 reports assigned to clusters and about five missed at cluster edges. A precision of 0.940 then implies that about 931 reports were assigned to clusters in total, of which roughly 56 were background reports; this agrees with the noise-rejection figure, since 91.0% of 620 background reports is about 564 rejected and therefore about 56 absorbed. These counts are derived by arithmetic from the reported rates and are approximate because the rates are rounded. They show that the errors are small in absolute terms: a handful of true reports are missed, and roughly one in sixteen reports inside a hotspot boundary is background.

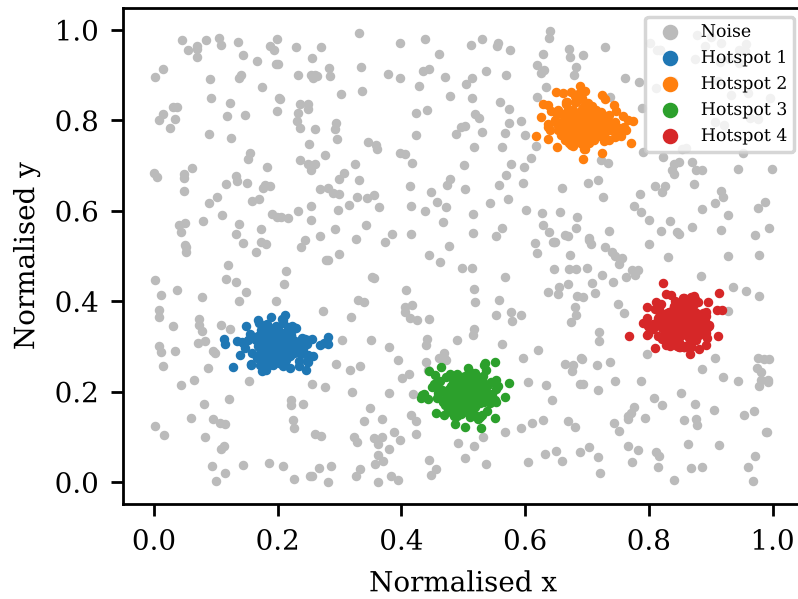


Fig. 3. Detected waste hotspots and rejected noise.

Table 7. Sensitivity of DBSCAN to ϵ and $\minPts$.

ϵ	$\minPts$	Clusters	Recall	Precision
0.015	10	4	0.901	0.974
0.015	15	4	0.830	0.977
0.015	30	4	0.543	0.982
0.03	10	4	0.997	0.934
0.03	15	4	0.994	0.940
0.03	30	4	0.988	0.953
0.06	10	11	1.000	0.675
0.06	15	5	1.000	0.826
0.06	30	4	1.000	0.874

Table 7 and Fig. 4 show that $\epsilon = 0.015$ with large $\minPts$ fragments hotspots and loses recall (down to 0.543), whereas $\epsilon = 0.06$ merges background noise into clusters and lowers precision (down to 0.675, with 11 clusters at $\minPts = 10$). The middle setting gives a good balance, but these values depend on the density of the simulated data, and real deployments should calibrate them per community.

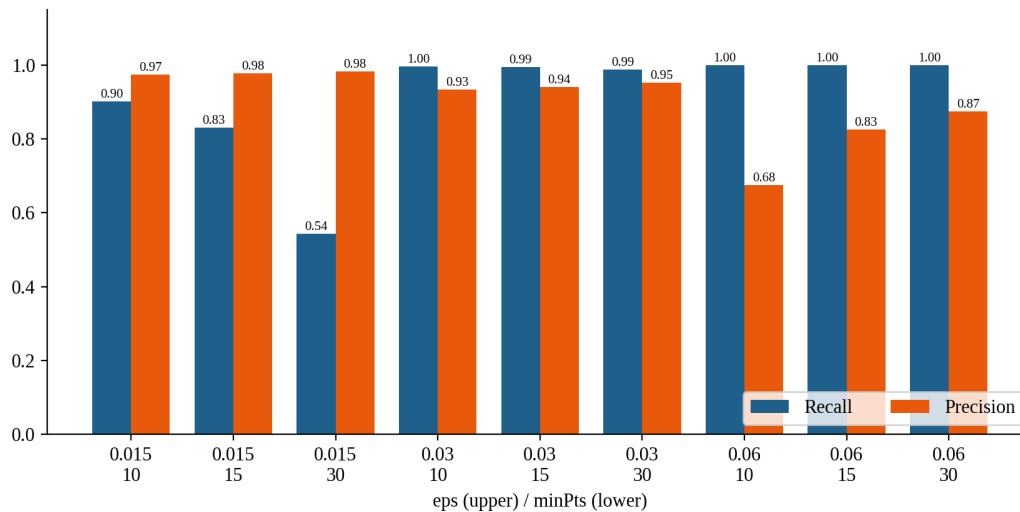


Fig. 4. Recall and precision of hotspot reports for the nine (eps, minPts) settings of Table 7.

Three patterns are visible in the sensitivity grid. First, the radius dominates: at the smallest radius, recall falls steeply as *minPts* rises, because at 0.015 only the densest cores of each hotspot satisfy the density criterion; at the largest radius, recall is perfect but precision suffers, because the neighbourhoods are large enough to link background points to hotspot cores. Second, *minPts* acts mainly as a stabiliser: at the largest radius, raising *minPts* from 10 to 30 raises precision from 0.675 to 0.874 and reduces the number of clusters from 11 to 4, since spurious clusters of background points need fewer than 30 neighbours to form but not more. Third, the $\text{eps} = 0.03$ row is robust: recall stays between 0.988 and 0.997 and precision between 0.934 and 0.953 across the three *minPts* values. A practical implication is that a deployment should first fix a radius corresponding to the physical scale of a hotspot and then use *minPts* to control how many reports are needed before a hotspot is declared.

4.3 Priority ranking

Top-5 wards by the priority score were 20, 1, 6, 10, 22, whereas ranking by the largest single risk gave 20, 6, 1, 14, 2. Only 3 of 5 wards are shared, so including exposure (population and vulnerability) and combining hazards changes the escalation order. Whether the combined ranking is closer to expert judgement could not be assessed without real expert data. The shared wards (20, 1 and 6) appear in a different order in the two lists, and wards 10 and 22, which are escalated by the priority score, are replaced by wards 14 and 2 under the single-risk ranking. Ward 20 is first in both lists, which indicates that it combines high risk with substantial exposure; the disagreement concerns the lower positions.

Weight sensitivity. We drew 500 weight vectors from a Dirichlet distribution centred on the default weights. The mean Jaccard overlap of the top-5 list with the default list was 0.75 (minimum 0.43), and the highest-priority ward was unchanged in 75% of runs (Fig. 5). The ranking is moderately stable, but a quarter of the runs changed the top ward, so the weights should be agreed with stakeholders.

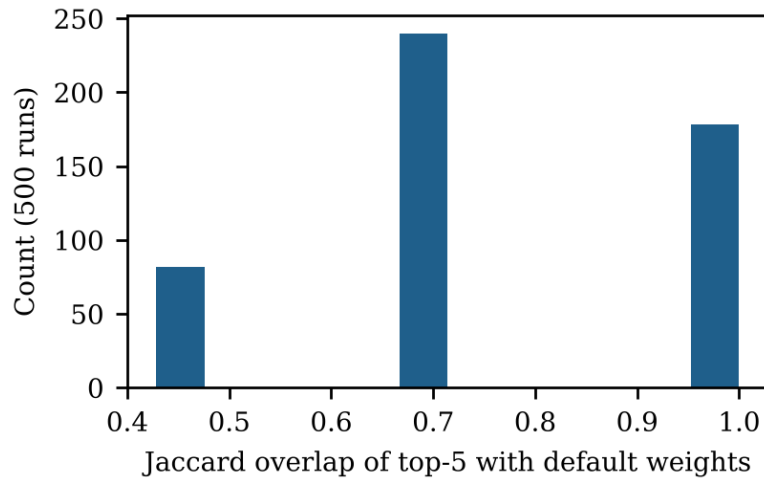


Fig. 5. Stability of the top-5 priority list under random weight perturbation.

The shape of Fig. 5 follows from the arithmetic of the Jaccard index. For two lists of five wards, sharing five wards gives a Jaccard overlap of $5/5 = 1.00$, sharing four gives $4/6 = 0.67$ and sharing three gives $3/7 = 0.43$; lower values would require at most two shared wards. The histogram therefore has three bars, and the minimum of 0.43 means that every perturbed list shared at least three wards with the default list. Reading the heights of the bars approximately from the figure (about 82, 240 and 178 runs, which sum to 500), the mean overlap is about 0.75, consistent with the reported value. About half of the runs changed exactly one ward in the top five, about a third left the list unchanged, and about a sixth changed two wards. No run changed more than two.

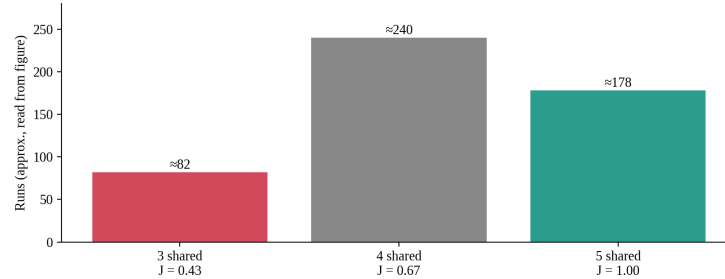


Fig. 6. Approximate distribution of top-5 overlap across the 500 weight perturbations (counts read from Fig. 5; Jaccard values follow from the number of shared wards).

4.4 What-if simulation

The baseline mean predicted flood probability was 0.259 (Fig. 7). Increasing drainage capacity by 25% and removing waste blockage reduced it to 0.155 (a reduction of 40.3%), while a 50% rainfall increase raised it to 0.365 (about 41% higher). The directions agree with expectations, which is a basic sanity check of the module. The percentages are those reported from the unrounded model output; computed from the rounded probabilities shown here they are 40.2% and 40.9%, which differ only in the last digit.

Two observations are worth making. First, the two scenarios have effects of similar size and opposite sign: in this simulation a modest improvement in drainage together with clearing waste blockage offsets roughly the effect of a 50% rise in rainfall. Second, the module reports a change in the mean over the evaluated records. A scenario may reduce the mean while leaving the highest-risk wards unchanged, so for operational decisions the distribution of changes across wards, and not only the mean, should be inspected. The what-if module is therefore best used to compare options (for example, drainage work versus waste clearing alone) rather than to predict absolute outcomes.

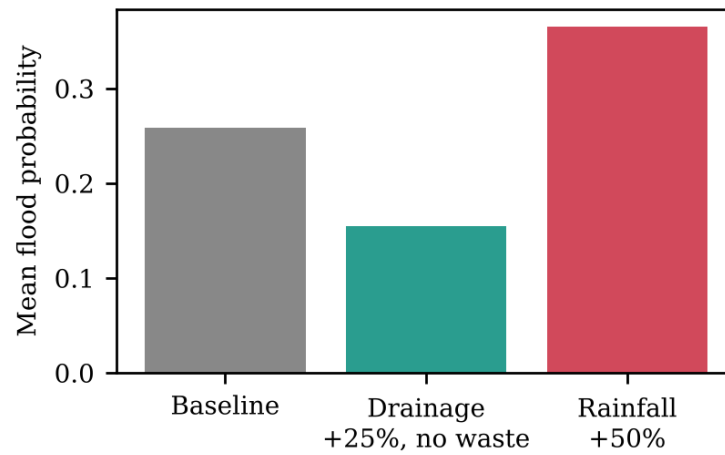


Fig. 7. Mean predicted flood probability under three scenarios.

4.5 Dashboard

Fig. 8 shows an illustrative mock-up of how the three components appear on the dashboard: a risk map with waste hotspots and flood- and water-risk locations, the ranked priority list, a community risk index and recommended actions. The values shown are illustrative only and are not outputs of a field deployment.



Fig. 8. Illustrative mock-up of the CommunityTwin dashboard.

5. Discussion

The experiments show that the three decision-support components work together on simulated data. Spatial clustering separated hotspots from noise, the priority score changed the escalation order relative to a naive ranking, and the what-if module produced plausible scenario effects. The following paragraphs relate these findings to the research questions and to practice.

RQ1: hotspot detection. On the simulated data, DBSCAN separated hotspots from scattered complaints with high recall and precision at a setting chosen a priori at the centre of the parameter grid. The sensitivity analysis shows, however, that the quality of the result depends on matching *eps* to the spatial scale of the hotspots and on using *minPts* to suppress spurious clusters. The simulated hotspots have equal size and spread, which is the easiest case for a single global density threshold. Real communities contain hotspots of different density, so a hierarchical variant (Campello et al. 2013) is a natural next step, and a deployment should include a calibration step in which staff inspect detected hotspots against known problem sites.

RQ2: prioritisation. The priority score changed which wards were escalated relative to ranking by the largest single risk. This is expected from the design: the score rewards wards that face several hazards and that have high exposure. Whether this is the right behaviour depends on policy. The weight-sensitivity analysis shows that the top-ranked ward was stable in three of four perturbed runs and that the top-5 list always retained at least three wards, so the ranking is informative but not insensitive. We therefore recommend that weights be agreed through a participatory process and that the dashboard show how a ward's rank would change under alternative weight sets, so that decisions that depend on contested weights can be identified.

RQ3: what-if analysis. The directions of the scenario effects agree with expectations, which is a necessary but not sufficient check. The magnitudes reflect the fitted model and the synthetic data. In particular, the waste-blockage index is a weak predictor in the simulation, yet scenario S1 sets it to zero together with a drainage increase, so the effect of S1 cannot be attributed to either change alone; separate scenarios would be needed to isolate them.

5.1 Practical implications

For a community starting to adopt such a system, the findings suggest the following. Citizen reporting is useful even when noisy, provided that a density criterion is applied before reports are allowed to influence decisions. The radius of the criterion should be set to the physical scale of a typical problem site. The priority score should be introduced as a discussion tool and its weights revisited after a trial period. And what-if outputs should be presented as comparisons between options with an explicit statement that they are statistical estimates.

5.2 Ethical and governance considerations

Citizen reports and images may contain personal information. A deployment should minimise the data it stores, remove faces and identifying text from images, and restrict access to the raw reports. Reporting rates differ between wards, so a ranking that depends on report counts may under-serve areas where residents report less. Monitoring the relation between report volume and rank, and complementing reports with sensor or field data, would help detect such effects. Finally, responsibility for decisions must remain with named officials; the system supports but does not replace their judgement.

5.3 Alternatives to density-based clustering

Several alternatives to DBSCAN could be used for hotspot detection, and the choice involves trade-offs. *Grid or zone counting* (counting reports per ward or per cell of a regular grid) is the simplest approach and matches how administrators already think, but it is sensitive to where cell boundaries fall and cannot follow elongated hotspots. *Kernel density estimation* produces a smooth intensity surface that is attractive on a map, but needs a bandwidth and a threshold to turn the surface into discrete hotspots, so it replaces two parameters with two others. *k-means* requires the number of hotspots to be fixed in advance and assigns every report, including isolated ones, to a cluster, which contradicts the aim of separating hotspots from one-off complaints. *Hierarchical density-based methods* (Campello et al. 2013) retain the noise concept and cope with varying density, at the price of somewhat greater complexity. DBSCAN is a reasonable first choice because its two parameters have direct physical meanings, but the comparison above indicates what should be tried if its single density threshold proves too restrictive in practice.

5.4 Computational considerations

DBSCAN, implemented with a spatial index, has an average running time of the order of $n \log n$ for n reports, although in the worst case it can approach n^2 . For the 1,500 reports used here this is negligible, and even for tens of thousands of reports per city the computation is modest. Because new reports arrive continuously, an operational system would re-cluster periodically (for example every few hours) or incrementally on the reports from the affected neighbourhood rather than on every new report. The priority score is a closed-form expression evaluated once per ward and the what-if module requires one pass of the flood model per scenario, so neither is a bottleneck. We did not measure running times for the components studied in this paper.

6. Limitations and Threats to Validity

The study has important limitations: (i) all data are synthetic, so results show feasibility, not field accuracy; (ii) the image classifier was not evaluated; (iii) priority weights were set by assumption, and a participatory process with local authorities is needed; and (iv) what-if results come from statistical models trained on correlations, not from hydrodynamic simulation, and should not be used as the sole basis for infrastructure investment.

Construct validity. The ‘true’ hotspots are Gaussian clouds of reports, and the background is uniform. Real report patterns are shaped by road networks, population density, reporting habits and campaigns, and may include repeated reports of the same heap by the same person. Recall and precision on simulated labels are therefore proxies for the quality of hotspot detection in practice. *Internal validity.* A single realisation of the data was used, and the headline parameter setting is one of nine in the grid; we did not repeat the experiment over multiple random seeds, so no confidence intervals are reported. *External validity.* Parameter values that work in normalised units on a unit square with 1,500 reports will not transfer directly to a real map of different extent, density and projection. *Conclusion validity.* The weight-sensitivity analysis uses a Dirichlet distribution centred on the defaults; a different perturbation distribution would give different stability figures. The agreement of the priority ranking with expert judgement could not be assessed.

7. Conclusions

This paper described the waste hotspot detection, priority engine and what-if simulation components of CommunityTwin. On simulated data, DBSCAN detected all four hotspots (recall 0.994, precision 0.940), the priority engine re-ordered wards compared with single-risk ranking, and the what-if module estimated a 40.3% drop in mean flood probability for a drainage and waste-clearing scenario. A sensitivity analysis showed that hotspot detection depends strongly on the neighbourhood radius and that the priority ranking is moderately but not completely stable under changes in weights.

Future work will evaluate a CNN-based waste image classifier, calibrate priority weights with stakeholders, and test the platform with real citizen reports in a pilot community. Further directions include hierarchical density-based clustering for hotspots of varying density, a time-aware severity score that decays as reports are

resolved, separate scenarios to isolate the effects of drainage and waste clearing, and a comparison of the ranking with expert judgement.

Appendix A: Reproducibility Summary

Table 8. Settings needed to reproduce the experiments (all taken from Section 4).

Component	Setting
Waste data	1,500 geotagged reports: 4 hotspots x 220 reports (Gaussian, SD 0.025, normalised units) + 620 uniform background reports; fixed random seed
Clustering	DBSCAN (scikit-learn), $\text{eps} = 0.03$, $\text{minPts} = 15$; sensitivity grid eps in $\{0.015, 0.03, 0.06\}$, minPts in $\{10, 15, 30\}$
Hotspot metrics	Recall and precision of hotspot reports; fraction of background reports rejected as noise; clusters found
Priority engine	25 simulated wards; weights water 0.35, flood 0.40, waste 0.25; exposure $0.6 + 0.25 \text{ pop} + 0.15 \text{ vul}$; levels by percentile
Weight sensitivity	500 Dirichlet-sampled weight vectors centred on defaults; Jaccard overlap of top-5; stability of top-ranked ward
What-if	Random forest flood model trained on 6,000 ward-day records; S1: drainage +25% and waste-blockage index = 0; S2: rainfall +50%

Appendix B: Design Options for the Severity Score

The experiments evaluate cluster detection, not the severity score, whose definition is left to configuration (Algorithm 1, step 5). Table 9 lists options that an implementation might consider; none is evaluated in this paper, and the choice should be made with local staff.

Table 9. Candidate designs for hotspot severity (not evaluated).

Option	Description	Consideration
Count only	Severity equals the number of open reports in the cluster	Simple; ignores how long the problem has persisted
Count x age weight	Reports weighted by how long they have been open (as in Algorithm 1)	Rewards persistent problems; needs a rule to stop old resolved reports counting
Count with decay	Older reports contribute less, so a hotspot fades when no new reports arrive	Avoids stale hotspots; may under-rank neglected sites with few repeat reports
Area-normalised	Severity divided by cluster area (report density)	Distinguishes compact heaps from sprawling sites; sensitive to eps
Category-weighted	Reports weighted by category (for example drainage-blocking waste counts more)	Links to flood risk; requires agreed category weights

Appendix C: Notation

Table 10. Notation used in this paper.

Symbol	Meaning
$\text{eps}, \text{minPts}$	DBSCAN neighbourhood radius and minimum number of reports for a core point
$s_{w,i}, s_{f,i}, s_{g,i}$	Normalised water, flood and waste risk scores of ward i
w_w, w_f, w_g	Weights of the water, flood and waste risk scores (defaults 0.35, 0.40, 0.25)
$\text{pop}_i, \text{vul}_i$	Normalised population and vulnerability index of ward i
P_i	Priority score of ward i , Eq. (1)
Jaccard overlap	$ A \cap B / A \cup B $ for two top-5 lists A and B

Appendix D: Proposed Pilot Evaluation

To move beyond feasibility, a pilot should compare the components with current practice. Table 11 lists measures that we propose; none has been applied in this study.

Table 11. Proposed outcome measures for a pilot deployment of the three components.

Component	Measure	Purpose
Hotspot detection	Share of detected hotspots confirmed by field inspection; share of known problem sites detected; reports needed before a hotspot is declared	Calibrate eps and minPts; estimate real precision and recall
Priority engine	Agreement between the ranking and an independent expert ranking; share of recommendations accepted by officials; ward rank versus report volume	Test weights; detect reporting bias
What-if module	Observed change in flooding after an intervention compared with the predicted change	Calibrate the effect magnitudes
Operations	Time from first report to action; collection frequency at hotspot sites; staff workload	Measure practical benefit

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None declared. This is a prototype study using simulated data.

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